Tarea 0: Setup & First Contact with All Three Datasets

IELE756 --- Preparación y Análisis de Datos

Group: 7

Team members: Macarena Fraga, Sebastian Bolomey

Region: Region Metropolitana

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Import: Pandas, Matplotlib

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
print("Hello, IELE756!")
print(f"pandas version: {pd.__version__}")
```

```
Hello, IELE756!
pandas version: 2.3.3
```

Part 1: Census 2024

1.1 Load

Loading the data

In [2]:
```
persona = pd.read_parquet(
    "personas_censo2024.parquet",
    columns=["region", "comuna", "sexo", "edad",
             "p27_nacionalidad", "p27_nacionalidad_rec",
             "escolaridad", "sit_fuerza_trabajo"],
)
```

1.2 Inspect

1.2.1 Number of rows and columns.

In [3]:
```
persona.shape
```

Out[3]:
```
(18480432, 8)
```

1.2.2 Data types

In [4]:
```
persona.dtypes
```

Out[4]:
```
region                    int32
comuna                    int32
sexo                      int32
edad                      int32
p27_nacionalidad          int32
p27_nacionalidad_rec      int32
escolaridad               int32
sit_fuerza_trabajo      float64
dtype: object
```

1.2.3 First 10 rows.

In [5]:
```
persona.head(10)
```

Out[5]:

	region	comuna	sexo	edad	p27_nacionalidad	p27_nacionalidad_rec	escolaridad
0	5	5802	2	80	1	1	17
1	5	5802	1	52	1	1	14
2	5	5802	2	45	1	1	12
3	5	5802	2	8	1	1	2
4	4	4303	1	69	1	1	12
5	4	4303	2	65	1	1	12
6	4	4303	1	58	1	1	15
7	11	11202	2	-66	-66	1	8
8	11	11202	1	-66	-66	1	5
9	11	11202	1	-66	-66	1	3

1.2.4 Null values

In [6]:
```
persona.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18480432 entries, 0 to 18480431
Data columns (total 8 columns):
 #   Column                Dtype
---  ------                -----
 0   region                int32
 1   comuna                int32
 2   sexo                  int32
 3   edad                  int32
 4   p27_nacionalidad      int32
 5   p27_nacionalidad_rec  int32
 6   escolaridad           int32
 7   sit_fuerza_trabajo    float64
dtypes: float64(1), int32(7)
memory usage: 634.5 MB
```

1.3 Filter for "Region Metropolitana"

In [7]:
```
my_region = persona[persona["region"] == 13]
print(f"Rows in my region: {len(my_region):,}")
```

```
Rows in my region: 7,400,741
```

1.4 First look at nationality

Percentage of foreigners in the selected region.

In [8]:
```
foreign = my_region["p27_nacionalidad_rec"] \
    .value_counts(normalize=True)
print(f"foreign-born: {foreign.get('Extranjero', 0):.1%}")
```

```
foreign-born: 0.0%
```

Part 2 - ENO: Notifiable Diseases

2.1 Load

In [9]:
```
eno = pd.read_csv(
    "20241218_base_eno_final.csv",
    sep=",", encoding="utf-8-sig", low_memory = False) #For small data like
print(f"Total rows: {len(eno):,}")
print(eno.columns.tolist())
```

```
Total rows: 333,300
['_id', 'etapa_clinica', 'region', 'seremi', 'nacionalidad', 'cie_10_diagnos
tico', 'diagnostico', 'Autoctono', 'anho_notificacion', 'ENO', 'pueblo_indig
ena', 'nombre_instruccion', 'presenta_sintomas', 'sintomas', 'consumo_agua_n
o_potable', 'antecedente_teniasis', 'imagenologia', 'manifestaciones', 'exis
te_coinfeccion', 'via_transmision', 'n_parejas_sexuales', 'uso_preservativo'
, 'pais_contagio', 'paciente_inmunosuprimido', 'vih', 'pareja', 'exantema_mp
ox', 'factores_exposicion', 'neumonia', 'contacto_aves', 'genotipo_1', 'recu
ento_linfocitos', 'clasificacion_final_vih', 'etapa_vih_sida', 'hospitalizac
ion', 'nombre_primer_sintoma', 'SeroGrupo', 'sexo', 'grupo_edad', 'codigo_co
muna_residencia']
```

Data type

In [10]:
```
eno.dtypes
```

Out[10]:
```
_id                          int64
etapa_clinica               object
region                      object
seremi                      object
nacionalidad                object
cie_10_diagnostico          object
diagnostico                 object
Autoctono                   object
anho_notificacion            int64
ENO                         object
pueblo_indigena             object
nombre_instruccion          object
presenta_sintomas           object
sintomas                    object
consumo_agua_no_potable     object
antecedente_teniasis        object
imagenologia                object
manifestaciones             object
existe_coinfeccion          object
via_transmision             object
n_parejas_sexuales          object
uso_preservativo            object
pais_contagio               object
paciente_inmunosuprimido    object
vih                         object
pareja                      object
exantema_mpox               object
factores_exposicion         object
neumonia                    object
contacto_aves               object
genotipo_1                  object
recuento_linfocitos         object
clasificacion_final_vih     object
etapa_vih_sida              object
hospitalizacion             object
nombre_primer_sintoma       object
SeroGrupo                   object
sexo                        object
grupo_edad                  object
codigo_comuna_residencia    object
dtype: object
```

10 most common diagnoses reported in this file

In [11]:
```
print(eno['diagnostico'].value_counts().head(10))
```

```
diagnostico
no aplica
69772
PAROTIDITIS, SIN COMPLICACIONES
47123
SIFILIS PRECOZ, LATENTE
24812
SIFILIS TARDIA, LATENTE
24269
INFECCION GONOCOCICA DEL TRACTO GENITOURINARIO INFERIOR SIN ABSCESO PERIURET
RAL O DE GLANDULA ACCESORIA    15399
HEPATITIS AGUDA TIPO A, CON COMA HEPATICO
13475
TOS FERINA DEBIDA A BORDETELLA PERTUSSIS
12762
SIFILIS, NO ESPECIFICADA
12615
INFECCION, GONOCOCICA, NO ESPECIFICADA
11695
SIFILIS SECUNDARIA DE PIEL Y MEMBRANAS MUCOSAS
10511
Name: count, dtype: int64
```

Regions in this file and number of data for each region.

In [12]:
```python
print(eno['region'].value_counts().head(17))
```

```
region
Región Metropolitana de Santiago                        126669
Región del Biobío                                        38592
Región de Valparaíso                                     36373
Región de Coquimbo                                       21373
Región de Antofagasta                                    18302
Región de Los Lagos                                      17829
Región de Tarapacá                                       13778
Región de la Araucanía                                   11593
Región del Libertador General Bernardo OHiggins          10765
Región del Maule                                         10128
Región de Atacama                                         7008
Región de Arica y Parinacota                              6541
Región de Los Ríos                                        5581
Región de Magallanes y la Antártica Chilena               3208
Región del Ñuble                                          2818
Región de Aysén del General Carlos Ibáñez del Campo       2690
Desconocido                                                 52
Name: count, dtype: int64
```

Percentage of ENO in the Metropolitan Region

In [13]:
```python
metrop = eno["region"] \
    .value_counts(normalize=True)
print(f"% ena in RM: {metrop.get('Región Metropolitana de Santiago', 0):.1%}
```

```
% ena in RM: 38.0%
```

2.2 Filter to region

Filtering data for the assigned region.

In [14]:
```
eno_region = eno[eno["region"] == "Región Metropolitana de Santiago"]
print(f"Rows in my region: {len(eno_region):,}".replace(",", "."))
```

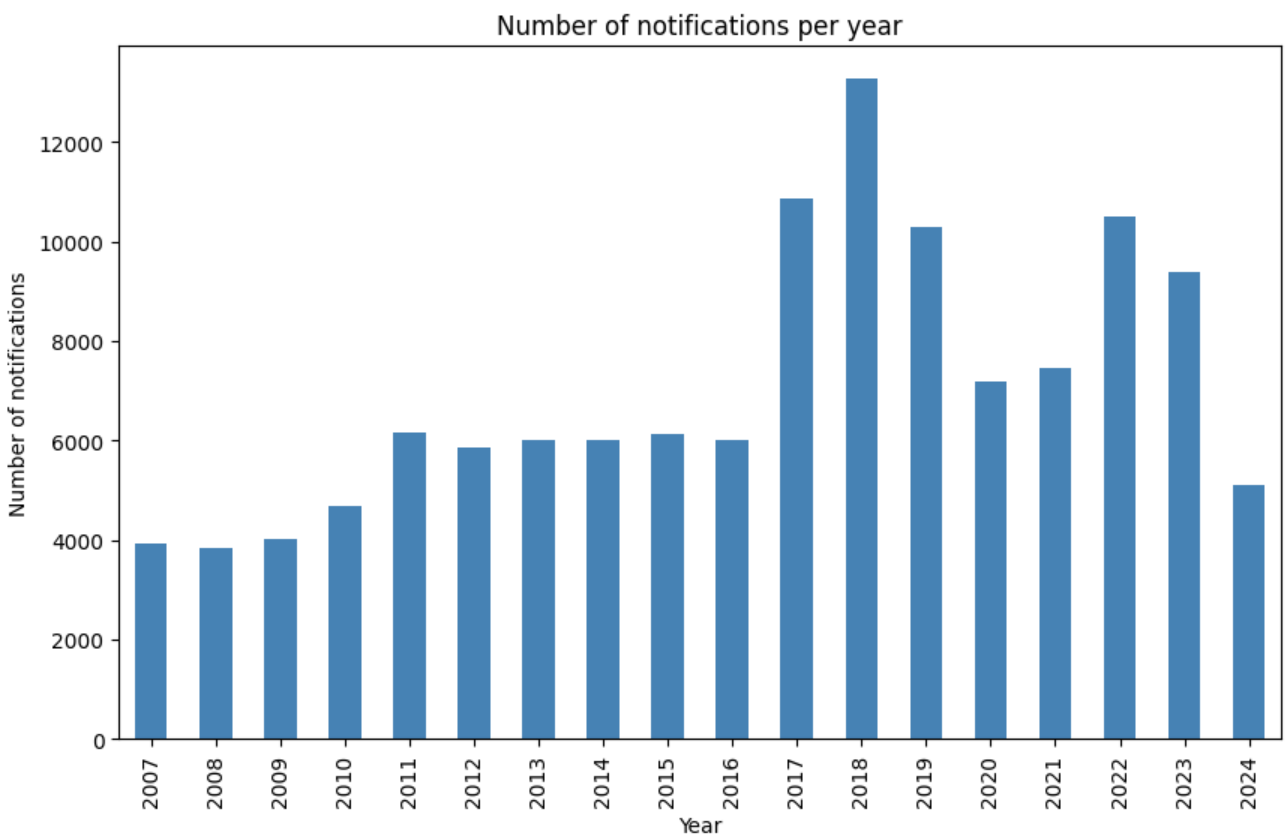
```
Rows in my region: 126.669
```

2.3 Notifications by year

Number of notifications per year

In [15]:
```
notifsxyear = eno_region["anho_notificacion"].value_counts().sort_index()
```

In [16]:
```
# Creating a bar chart
notifsxyear.plot(kind="bar", figsize=(10, 6), color="steelblue")

# Labels and title
plt.title("Number of notifications per year")
plt.xlabel("Year")
plt.ylabel("Number of notifications")

#Display chart
plt.show()
```



2.4 - Top 5 diseases

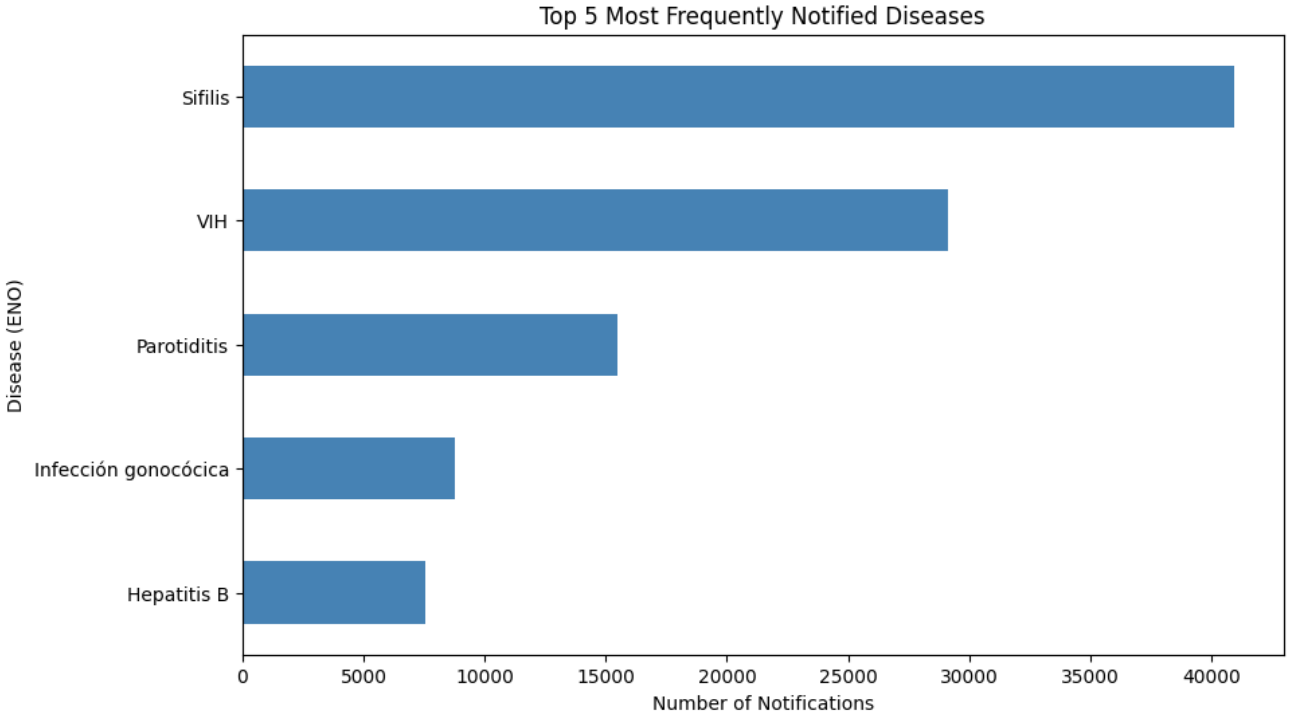
5 most frequently notified diseases in Region Metropolitana

In [17]:
```
top_5_diseases= eno_region["ENO"].value_counts().head(5)
print(top_5_diseases)
```

```
ENO
Sifilis                 40928
VIH                     29097
Parotiditis             15493
Infección gonocócica     8766
Hepatitis B              7565
Name: count, dtype: int64
```

Displaying the top 5 most frequently notified diseases in Region Metropolitana as an horizontal bar chart

In [18]:
```
# Most frequent disease appear at the top
top_5_diseases = top_5_diseases.sort_values(ascending = True)

# For horizontal bar chart
top_5_diseases.plot(kind="barh", figsize=(10, 6), color="steelblue")

# Labels and title
plt.title("Top 5 Most Frequently Notified Diseases")
plt.xlabel("Number of Notifications")
plt.ylabel("Disease (ENO)")

#Display chart
plt.show()
```



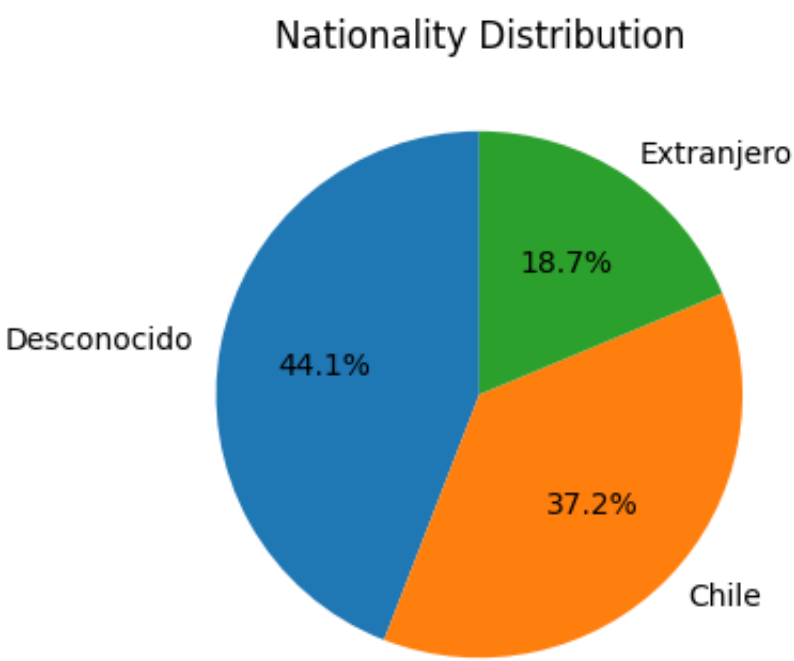
2.5 Nationality distribution

In [19]:
```
nat_dist = eno_region["nacionalidad"].value_counts()
print(nat_dist)
```

```
nacionalidad
Desconocido    55863
Chile          47093
Extranjero     23713
Name: count, dtype: int64
```

With a Pie Chart

In [20]:
```
#Creating pie chart
nat_dist.plot(
    kind='pie',
    figsize=(4, 4),          # Makes the chart a big square
    autopct='%1.1f%%',       # Calculates and shows the percentages
    startangle=90,           # Rotates the first slice to start at the top
    title="Nationality Distribution"    # Adds a title
)

plt.ylabel("")
plt.show()
```



Part 3: GRD - Hospital Discharges

3.1 Load year 2024

In [21]:
```
import zipfile

cols = ["COMUNA", "NACIONALIDAD", "SEXO", "DIAGNOSTICO1",
        "FECHA_INGRESO", "FECHAALTA",
        "IR_29301_SEVERIDAD", "IR_29301_COD_GRD"]

with zipfile.ZipFile("GRD_PUBLICO_2024.zip") as z:
    with z.open("GRD_PUBLICO_2024.txt") as f:
        grd = pd.read_csv(f, sep="|", usecols=cols,
                          low_memory=False, encoding="latin1")

print(f"Total discharges: {len(grd):,}")
```

```
Total discharges: 1,085,813
```

3.2 Filter to your region

In GRD there is no region column. Instead, COMUNA contains the commune name in uppercase (e.g., "IQUIQUE").

But first we have to know what comunas are we working with. Therefore we use the comunas in the selected region.

In [22]:
```
comunas = my_region['comuna'].unique().tolist()
print(comunas)
print(len(comunas))
```

```
[13124, 13101, 13603, 13501, 13107, 13119, 13126, 13123, 13106, 13401, 13120
, 13130, 13503, 13605, 13113, 13127, 13110, 13201, 13602, 13302, 13301, 1311
4, 13111, 13103, 13128, 13115, 13102, 13132, 13108, 13105, 13125, 13122, 131
31, 13129, 13104, 13121, 13117, 13404, 13604, 13112, 13118, 13402, 13403, 13
109, 13116, 13601, 13203, 13505, 13303, 13502, 13202, 13504]
52
```

The comunas are displayed as their territory name, so we first have to change them using the dictionary given by the Censo 2024.

First we upload the given dictionary on xlsx.

In [23]:
```
# 1. Read the Excel file
comunas_df = pd.read_excel('diccionario_variables_censo2024.xlsx',sheet_name

# Printing the columns so you can see the exact names
print(comunas_df.columns.tolist())
```

```
['Código territorial', 'División Política Administrativa', 'Territorio']
```

Last, we create a dictionary so it can translate the comunas code number in our region into a string:

In [24]:
```
# 2. Creating dictionary
codigo_a_nombre = dict(zip(comunas_df['Código territorial'], comunas_df['Ter

# 3. Add the translated names directly to your original DataFrame
my_comunas = [codigo_a_nombre.get(codigo, "Desconocido") for codigo in comur

# Check the results
print(my_comunas)
print(len(my_comunas))
```

```
['Pudahuel', 'Santiago', 'Isla de Maipo', 'Melipilla', 'Huechuraba', 'Maipú'
, 'Quinta Normal', 'Providencia', 'Estación Central', 'San Bernardo', 'Ñuñoa
', 'San Miguel', 'Curacaví', 'Peñaflor', 'La Reina', 'Recoleta', 'La Florida
', 'Puente Alto', 'El Monte', 'Lampa', 'Colina', 'Las Condes', 'La Granja',
'Cerro Navia', 'Renca', 'Lo Barnechea', 'Cerrillos', 'Vitacura', 'Independen
cia', 'El Bosque', 'Quilicura', 'Peñalolén', 'San Ramón', 'San Joaquín', 'Co
nchalí', 'Pedro Aguirre Cerda', 'Lo Prado', 'Paine', 'Padre Hurtado', 'La Pi
ntana', 'Macul', 'Buin', 'Calera de Tango', 'La Cisterna', 'Lo Espejo', 'Tal
agante', 'San José de Maipo', 'San Pedro', 'Tiltil', 'Alhué', 'Pirque', 'Mar
ía Pinto']
52
```

In [25]:
```
my_comunas = ["COMUNA1", "COMUNA2", ...]  # your region's comunas
grd_region = grd[grd["COMUNA"].isin(my_comunas)]
print(f"Discharges in my region: {len(grd_region):,}")
```

```
Discharges in my region: 0
```

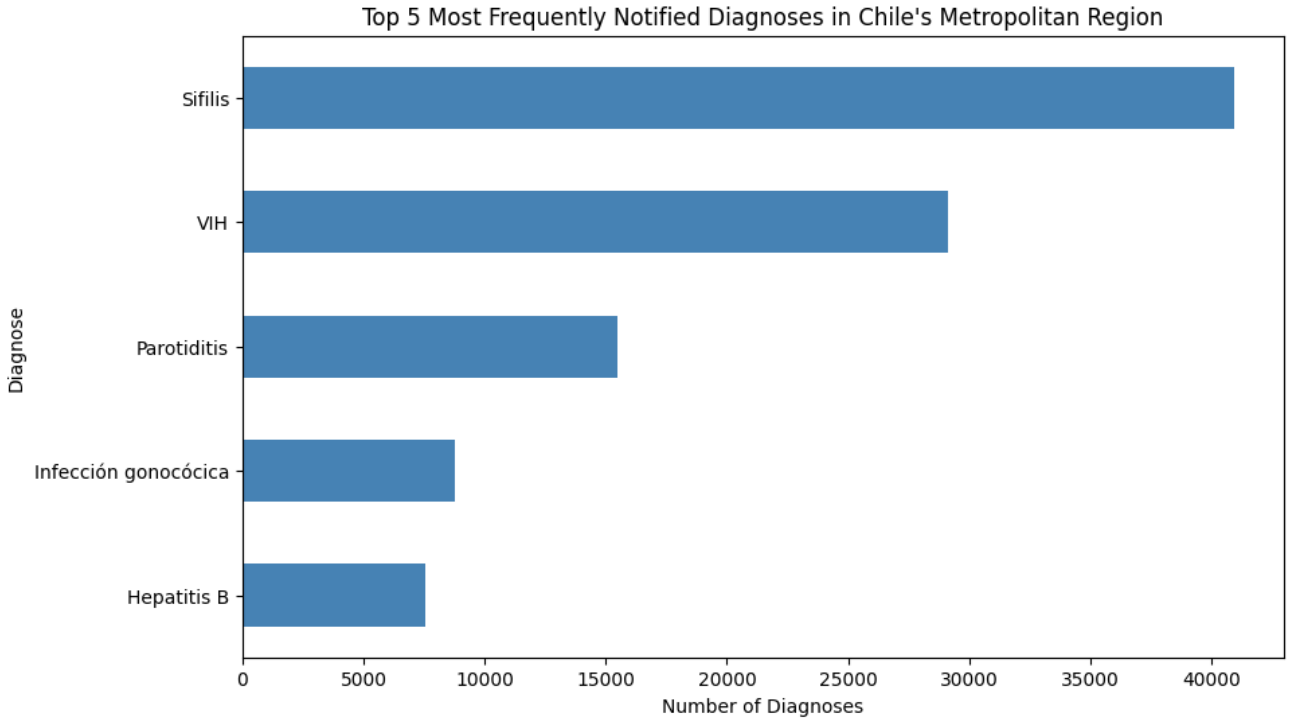
3.3 Join with CIE-10

The DIAGNOSTICO1 column contains ICD-10 codes (e.g., "J18.9"). To get human-readable names, join with the lookup table:

In [26]:
```
cie10 = pd.read_excel("CIE-10.xlsx",
                      sheet_name="CIE 10")

grd_region = grd_region.merge(
    cie10[["Código", "Descripción", "Capítulo"]],
    left_on="DIAGNOSTICO1", right_on="Código",
    how="left")
```

3.4 Top 5 diagnoses

Show the 5 most common diagnoses (by Descripción) in your region:

In [27]:
```
top_5_diagnoses = grd_region["Descripción"].value_counts().head(5)

# Most frequent diagnoses appear at the top
top_5_diagnoses = top_5_diseases.sort_values(ascending = True)

# For horizontal bar chart
```

```python
top_5_diagnoses.plot(kind="barh", figsize=(10, 6), color="steelblue")

# Labels and title
plt.title("Top 5 Most Frequently Notified Diagnoses in Chile's Metropolitan
plt.xlabel("Number of Diagnoses")
plt.ylabel("Diagnose")

# Display chart
plt.show()
```